

Chapter 13: High-Yield NBDHE Study Guide

Dental Biomaterials & Clinical Applications (English)

Dental Biomaterials & Clinical Applications

1. Impression Materials

- **Hydrocolloids:**
 - **Alginate (Irreversible Hydrocolloid):** Sets by chemical reaction. Susceptible to **syneresis** (loss of water / shrinkage) and **imbibition** (absorption of water / expansion). Pour within 10–15 minutes.
 - **Agar (Reversible Hydrocolloid):** Changes state via temperature changes (gel $\rightarrow$ sol).
- **Elastomers:**
 - **Vinyl Polysiloxane (VPS / Addition Silicone):** Highest dimensional stability; no volatile by-product; mixed through automix nozzle.
 - **Polyether:** Excellent hydrophilicity; rigid setting; absorbs water if stored in humid environments.
 - **Polysulfide (Rubber Base):** Smelly (sulfur by-product); pour promptly.

2. Direct Restorative Materials & Adhesion

- **Dental Amalgam:**
 - Alloy powder (Silver, Tin, Copper, Zinc) + Liquid Mercury (Hg).
 - High copper amalgams ($\geq 12\%$) eliminate the corrosive **γ_2 (Sn-Hg)** phase.
 - Trituration: Mechanical mixing in pre-dosed capsules minimizes mercury vapor release (OSHA limit: 0.05 mg/m³ for 40-hr week).
 - Creep: Slow dimensional distortion under sustained occlusal load.
- **Composite Resins:**
 - Composition: Resin matrix (Bis-GMA, UDMA, TEGDMA) + Inorganic filler particles (silica, zirconia) + **Silane coupling agent**.
 - Polymerization shrinkage: Causes marginal leakage, post-operative sensitivity, and recurrent decay. Incremental placement (≤ 2 mm) counters C-factor stress.
- **Adhesive Systems:**
 - **Etch-and-Rinse (Total-Etch):** 35%–37% Phosphoric acid dissolves hydroxyapatite, creating microporosities in enamel and exposing collagen fibrils in dentin.
 - **Hybrid Layer:** Infiltration of hydrophilic resin monomers (HEMA) into demineralized dentin collagen meshwork forming micromechanical resin tags.

3. Dental Cements & Liners

- **Zinc Phosphate:** Exothermic setting reaction $\rightarrow$ mix on a **chilled glass slab** in small increments over a wide area. Low initial pH causes pulpal irritation.
- **Zinc Oxide-Eugenol (ZOE):** Soothing sedative effect on pulp (obtundent); **contraindicated under composite resins** because eugenol inhibits free-radical polymerization.
- **Glass Ionomer (GI / RMGI):** Releases fluoride; bonds chemically to calcium in enamel/dentin via carboxylate groups; thermal expansion coefficient matches natural tooth structure.

4. Gypsum Products & Ceramics

- **Gypsum (Calcium Sulfate Dihydrate $\rightarrow$ Hemihydrate):**

- Type II: Model Plaster (large, irregular particles, highest water-to-powder ratio).
- Type III: Dental Stone (yellow stone, study models).
- Type IV: High-Strength Die Stone (dense, cuboidal particles, lowest expansion).

- **Dental Ceramics:**

- Glass-matrix ceramics (Feldspathic, Lithium Disilicate/E.max) $\rightarrow$

etched with **Hydrofluoric (HF) acid** and silanated.

- Polycrystalline ceramics (Zirconia) $\rightarrow$

extremely high flexural strength (>900 MPa); bonded using **MDP primers** and sandblasting (cannot be etched with HF acid).

5. Dental Implants & Tissue Engineering

- **Implants:** Made of commercially pure **Titanium** or titanium alloy; osseointegration occurs with biocompatible titanium oxide (TiO_2) surface. Clean with non-metallic plastic, graphite, or titanium scalers to prevent scratching.

- **Tissue Engineering Triad:** **Cells + Signals** (growth factors) + **Scaffolds** (e.g., collagen, PLA, bioglass).

- **Bone Grafts:**

- **Autograft:** Bone from the patient's own body (gold standard, osteogenic).
- **Allograft:** Bone from human cadaver donors.
- **Xenograft:** Bone from another species (e.g., bovine).
- **Alloplast:** Synthetic bone substitute (hydroxyapatite, beta-TCP).